

Project #9: EMPOWERED GRASP

Myoelectric Smart Embedded Prosthetic Hand with Intelligent Capabilities

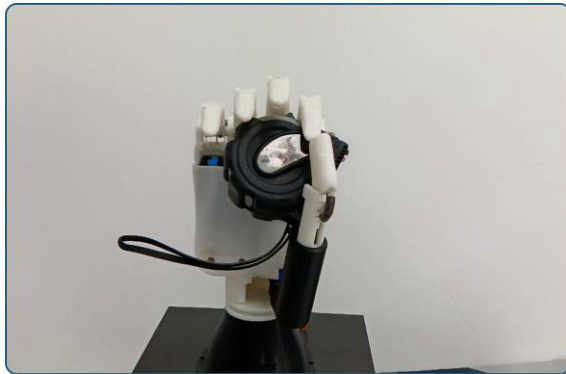
Hardware • Embedded Systems • Intelligent Control • Mobile/PC Application

Project Overview

This graduation project is a direct continuation of the prosthetic-hand platform developed during the previous academic year. It will transform the existing five-finger mechatronic prototype into a compact, wearable and intelligent assistive device for people with transradial (below-elbow) limb loss or congenital upper-limb difference.

The upgraded platform integrates an improved mechanical hand and actuator transmission, surface electromyography (EMG), an embedded real-time controller, machine-learning-based grasp recognition, sensor feedback, a rechargeable battery with battery-management system (BMS), and wireless tuning through mobile, tablet and PC software.

Aligned with Egypt Vision 2030 and equitable healthcare, the project emphasizes locally developed assistive technology that is intuitive, safe, maintainable and affordable, while providing a research platform for advanced prosthetic control and rehabilitation engineering.



Existing five-finger myoelectric prosthetic-hand prototype.

Main Objectives for the New Project Phase

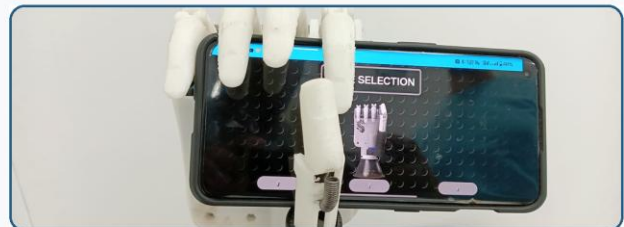
- Assess and enhance the previous-years prototype rather than rebuilding the system from the beginning.
- Reduce weight, backlash and bulk while improving finger transmission, thumb opposition and grasp reliability.
- Develop EMG acquisition, Myo Plus/adaptive pattern-recognition control and embedded safety supervision.
- Integrate position, motor-current and optional fingertip force/tactile feedback for closed-loop grasping.
- Provide BMS-protected power, Bluetooth/Wi-Fi connectivity and mobile/tablet/PC calibration software.
- Validate the system using representative daily-living tasks and measurable engineering performance criteria.

Previous-Year Prototype: Demonstrated Daily-Living Functions



Power / cylindrical grasp

Tape measure and everyday objects



Smartphone interaction

Mode selection and digital-device use



Precision / tool grasp

Stable pen and soldering-tool handling



Carry / handle grasp

Backpack and handle load support

Figure 1. Functional evidence from the existing prototype: power grasp, smartphone interaction, precision/tool handling and load carrying.



Previous-Year Project Videos

The linked videos are retained as the baseline reference for the previous design, fabrication, software interface and functional demonstrations. [Video 1](#) | [Video 2](#)



1. Integrated Project System, Software and Tuning Architecture

The following system-level layout consolidates the planned hardware, embedded control, intelligent EMG processing, functional grasp modes, wireless communication and user-facing tuning software. It also shows how the mobile and PC interfaces support calibration, mode selection, grip-force and speed adjustment, battery monitoring, diagnostics and performance review.



Empowered Grasp

Myoelectric Smart Embedded Prosthetic Hand with Intelligent Capabilities

Hardware • Embedded Systems • Intelligent Control • Mobile App

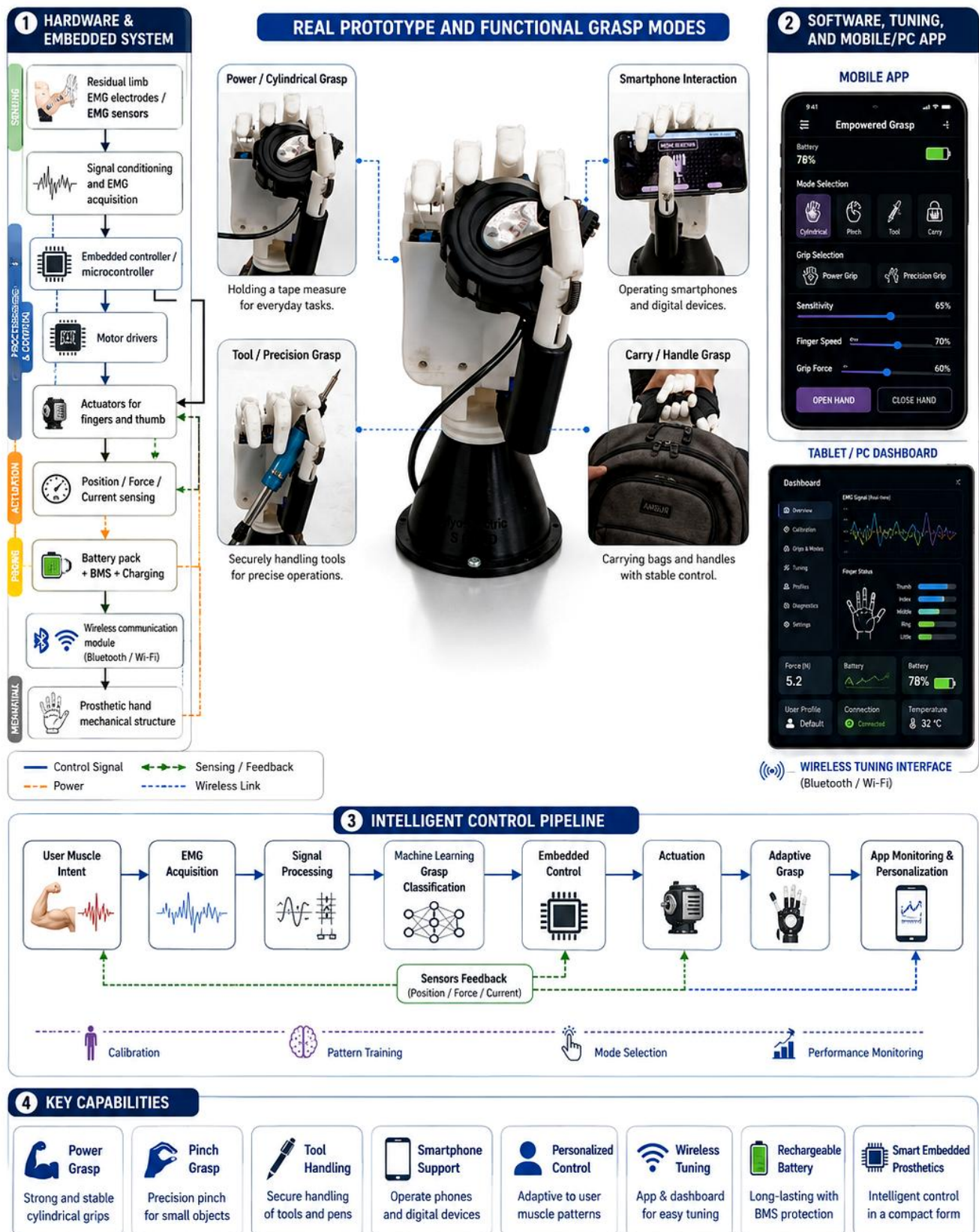


Figure 2. Complete project architecture linking the physical prosthetic hand, EMG sensing, embedded intelligence, closed-loop feedback, wireless tuning and mobile/PC applications.

2. Technical Scope, Work Packages and Validation Plan

The new project phase will be implemented as an integrated hardware-software-embedded development program. Mechanical, electronic, control, power and application requirements will be treated concurrently so that signal quality, latency, safety, energy consumption, communication reliability and user experience are considered from the beginning.

Work Package	Principal Components	Planned Development and Outputs
Mechanical & actuation	Anthropomorphic five-finger structure; thumb opposition; compact motors; tendons/gears; palm, wrist and socket interfaces.	Redesign for lower weight and volume; reduce backlash; increase range of motion and grasp conformity; improve serviceability and robustness.
EMG & feedback sensing	Surface EMG electrodes and analogue front end; finger position; motor current; optional fingertip force/tactile sensing.	Improve signal-to-noise ratio and artefact rejection; calibrate sensors; detect contact, slip, stall and overload; support proportional and closed-loop control.
Embedded electronics	Low-power microcontroller/edge processor; motor drivers; real-time firmware; watchdog; local memory; Bluetooth/Wi-Fi.	Implement deterministic acquisition and control loops, grasp state machine, safety limits, logging, communication protocol and diagnostic functions.
Power & BMS	Rechargeable battery, charging electronics, protection, voltage/current/temperature monitoring and power distribution.	Estimate remaining energy; prevent under/overvoltage, overcurrent and overheating; evaluate operating time and energy per grasp cycle.
AI, Myo Plus & software	EMG features and classifier; confidence logic; mobile/tablet/PC application; profiles and data storage.	Train personalized grasp models; provide calibration, pattern training, mode selection, sensitivity, speed and force tuning, live plots, reports and clinician/user review.

Personalized Myoelectric Control, Safety and Software Workflow

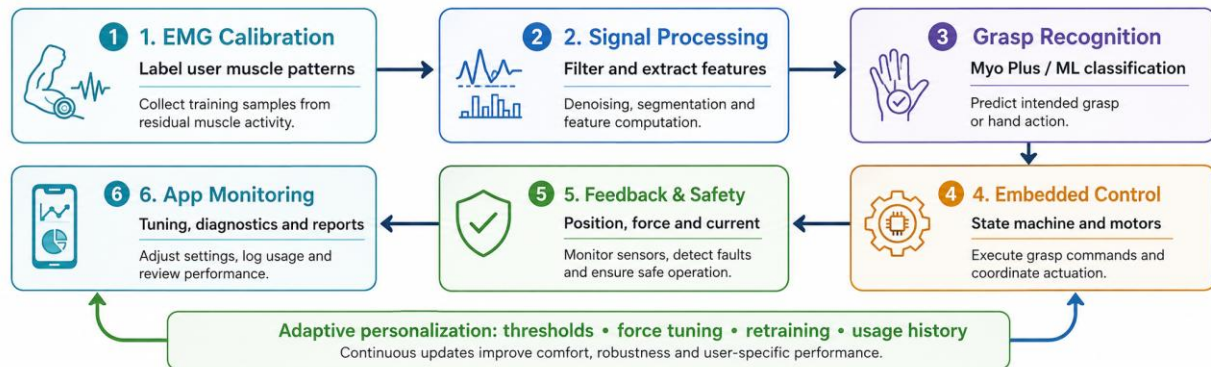


Figure 3. End-to-end personalized myoelectric operation, intelligent grasp selection, embedded execution, sensor-based safety and software monitoring.

Development Methodology

- Benchmark the previous prototype and define requirements for mass, dimensions, grasp speed, payload, battery life and safety.
- Optimize mechanical parts and transmissions using CAD, rapid prototyping and actuator/bench testing.
- Develop the EMG board, sensor interfaces, motor drivers, BMS and real-time embedded firmware.
- Collect labelled EMG data and compare conventional classifiers with lightweight edge-deployable machine-learning models.
- Integrate the mobile/tablet/PC application and complete subsystem, integrated and daily-living task tests.

Evaluation Metrics

- EMG classification accuracy, confidence, response latency and calibration time.
- Grasp completion time, repeatability, object-holding success and unintended-command rate.
- Safe payload, contact force, current margins, stall/overload detection and emergency release.
- Battery runtime, energy per grasp, temperature and communication reliability.
- Application usability, profile personalization, comfort and perceived control.

Mobile/PC Tuning and Monitoring Functions

- User registration, EMG calibration and labelled pattern-training sessions.
- Selection of power, pinch, tripod, lateral/key, tool and custom grasp modes.
- Sensitivity, confidence threshold, finger speed and grip-force adjustment.
- Manual open/close testing, real-time EMG visualization and finger-status display.
- Battery state, temperature, connection status, faults, usage history and report export.

Expected Deliverables

- Enhanced five-finger prosthetic-hand prototype and wearable integration concept.
- Embedded controller, EMG/sensor electronics, motor drivers and protected BMS.
- Real-time EMG/ML grasp-control firmware with closed-loop safety functions.
- Mobile/tablet/PC calibration, tuning, monitoring and reporting application.
- Test dataset, engineering validation report, user manual and demonstration video.

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